SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA51
COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS: **Total Marks: 50**

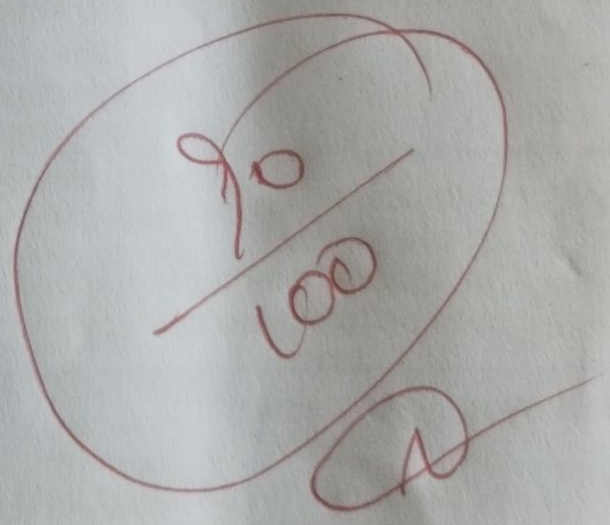
Annexure A

Code of Conduct:

I D·Naveen *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

D·Naveen



Analyze why hybrid encryption is more efficient
Justify its use in modern secure communication systems

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20% 20	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20% 20	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25% 20	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process

Accuracy of Calculation	20% 20	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15% 10	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

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1) RSA Key Generation and security Analysis:

Given: $p=17$, $q=19$, $e=5$, $M=7$.

(a) Calculate n and $\phi(n)$.

$$n = p \times q = 17 \times 19 = 323$$

$$\phi(n) = (p-1)(q-1)$$

$$= (17-1)(19-1) = 16 \times 18$$

$$\boxed{\phi(n) = 288}$$

(b) Calculate the private key d.

The private key satisties:

$$ed = 1 \pmod{\phi(n)}$$

$$5d = 1 \pmod{288}$$

using the extend Eculidean algorithm:

$$288 = 5(57) + 3$$

$$5 = 3(1) + 2, \quad 3 = 2(1) + 1$$

Therefore,

$$1 = 3 - (288 - 5(57)) - (288 - 5(57) - 5)$$

Equivalently, $1 = 116(5) - 2(288)$

$$\boxed{d = 173}$$

Correction: Since $5(173) = 865 = 3(288) + 1$, the correct private key is

$d = 173$

Public key: $(e, n) = (5, 323)$

Private key: $(d, n) = (173, 323)$

(c) Encrypt the message $M = 7$.

RSA encryption formula:

$$C = M^e \bmod n$$

$$C = 7^5 \bmod 323$$

$7^2 = 49$, $7^4 = 2401 = 140 \pmod{323}$

$C = (140 \times 7) \bmod 323$

$$\boxed{C = 11}$$

(d) Decrypt and verify:-

$$M = C^d \bmod n = 11^{173} \bmod 323$$

using modular exponentiation, the result is:

$$\boxed{M = 7}$$

The decrypted message equals the original message so encryption and decryption are verified.

1) Impact of Small prime numbers on RSA Security:-

* Small primes produce a modulus n.
* An Attacker can factor n more easily to find p and q.
* once the factors are known, the attacker can calculate $\phi(n)$ and derive the private key.
* Therefore, real RSA systems use sufficiently large primes and secure key-generation methods.

2) ElGamal Encryption Analysis:-

Given $p = 23$, $g = 5$, $x = 7$, $k = 3$, $m = 10$.

(a) Calculate the public key:-

The public key component is:

$$y = g^x \bmod p.$$

$$y = 5^7 \bmod 23$$

Since $5^2 = 25 = 2$, $5^4 = 4$, and $5^7 = 4 \times 2 \times 5 = 40$.

$$\boxed{y = 17}$$

public key: $(p, g, y) = 23, 5, 17$.

private key: $x = 7$.

(b) Encrypt the message

Calculate the ciphertext component

$$C_1 = g^k \bmod p$$

$$C_1 = 5^3 \bmod 23 = 10$$

Calculate the shared value:

$$K = y^k \bmod p$$

$$K = 17^3 \bmod 23 = 14.$$

Calculate the second ciphertext component

$$C_2 = m \times K \bmod p$$

$$C_2 = 10 \times 14 \bmod 23$$

$$\boxed{C_2 = 2}$$

Thus, the ciphertext pair is

$$\boxed{(C_1, C_2) = (10, 2)}$$

(c) decrypt the message:-

Calculate the shared value using the private key.

$$K = C_1^x \bmod p = 10^7 \bmod 23 = 14.$$

The modular inverse of 14 modulo 23 is 5, because $14 \times 5 \bmod 23 = 1$

decryption formula:-

$$M = C_2 k^{-1} \bmod p$$

$$M = 2 \times 5 \bmod 23$$

$$\boxed{M = 10}$$

The original message is recovered successfully.

(d) How randomness improves ElGamal security:-

* A fresh random values k should be selected for each encryption message.
* Different values of k produce different ciphertexts for the same message.
* This makes it harder for attacker to identify repeated message.
* Reusing the or the exposing k can compromise security.

3. RSA Digital signature verification:-

Given $p=11$, $q=17$, $e=7$, $m=8$

(a) calculate n and $\phi(n)$.

$$n = pq = 11 \times 17 = 187.$$

$$\phi(n) = (11-1)(17-1) = 10 \times 16$$

$$\boxed{\phi(n) = 160}$$

(b) Find the private key d

$$ed = 1 \bmod 160.$$

$$7d = 1 \bmod 160$$

Since $7 \times 23 = 161$

$\boxed{d = 23}$

public key: 7, 187.

private key: 23, 187.

(c) Generate the digital signature:-

The signature is calculated using:

$$S = m^d \bmod n$$

$$S = S^{23} \bmod 187.$$

using modular exponeration:

$\boxed{S = 83}$

(d) verify the signature:-

The reciver verifies the signature using the public key.

$$m' = S^e \bmod n$$

$$m' = 83^7 \bmod 187$$

$\boxed{m' = 8}$

Since $m' = M = 8$, the digital signature is valid.

(e) How digital signature ensure non-repudiation:-

* The Sender generates the signatures using the private key.
* The reciver verifies it using the Sender's public key.
* The valid signature supports message integrity and evidence of the signature's involvement.